
Session 2 (2h) — 3D Features & Part Modeling

Core concepts (1h)

- **Extrude** — turns a closed 2D sketch profile into a solid by pushing it along an axis. Two directions matter: distance and symmetric/one-side.
- **Revolve** — spins a profile around an axis (good for round parts: knobs, bushings).
- **Fillet / Chamfer** — rounds or bevels edges. Real printed parts almost always need small fillets on external edges (sharper edges are weaker and print worse).
- **Shell** — hollows out a solid, leaving a wall thickness — this is how you make a housing instead of a solid block.
- **Feature timeline** — Fusion records every step as a history tree at the bottom of the screen. You can go back and edit any earlier feature; everything after it updates automatically. This is what “parametric” means, and it’s covered more in Session 4.

On-site task link: 부품 모델링 (part modeling)

The internship’s on-site work explicitly includes modeling parts and housings — this session’s practice is a direct rehearsal of that task.

Hands-on practice (1h)

1. Extrude the rectangle sketch from Session 1 into a 20mm-tall block.
2. Add a fillet (2mm) to the four vertical edges.
3. Shell the block (2mm wall thickness), leaving the top face open — this is now a basic enclosure body.
4. On a new sketch (on top of a cylinder), practice Revolve: sketch a simple L-shaped profile and revolve it 360° to make a knob or standoff shape.
5. Save the file — you’ll extend this enclosure in Session 3–5.

Checkpoint

- ☐ Can extrude a sketch into a solid and explain symmetric vs one-sided extrude
- ☐ Produced a shelled enclosure body with filleted edges
- ☐ Used Revolve at least once and understands when to prefer it over Extrude